

Ahmad Maaz

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EXPERIENCE

University of Regina

Jun 2025 – Sep 2025

Mitacs Globalink Research Fellow, Generative AI & HCI

Regina, SK, Canada

- Built an end-to-end **RAG ingestion pipeline** (n8n, Docker, Supabase, PostgreSQL) integrating the **Google Drive API** to auto-pull new files and extract text across PDF, HTML and OCR-required formats into a 44-chapter knowledge base.
- Architected a self-correcting **multi-agent loop** of generator, evaluator and tester agents that automatically re-routed failing answers for regeneration until they passed quality checks.
- Designed a six-category **LLM evaluation suite** and benchmarked prompt variants against it: explicit structural rules beat few-shot examples on output consistency, while hallucination on out-of-scope questions persisted across every variant.
- Documented test methodology, results and known limitations so the lab could continue the work; delivered the system solo, reporting directly to the supervising professor.

Motive Technologies, Inc.

Dec 2025 – Jul 2026

Technical Support Engineer, AI Hardware Validation

United States (Remote)

- Led technical validation for **AI-powered dashcams**: verified AI-detected safety events, diagnosed false-positive patterns and escalated confirmed model failures to engineering, feeding into alert accuracy and product reliability.
- Diagnosed and resolved hardware, firmware and connectivity faults across Motive's device line, **Vehicle Gateways (FMCSA-registered ELDs)**, **AI Dashcams** and **Asset Gateways**, keeping regulated fleets compliant and operational.

University of Europe for Applied Sciences

Jun 2024 – Sep 2024

Generative AI Research Intern

Potsdam, Germany (Remote)

- Built a **RAG-based legal assistant** over three German legal corpora using a quantized **Llama 2 7B** model, ChromaDB and LangChain.
- Benchmarked generated answers against human legal expert responses using **BLEU and ROUGE**, with a 72.36s average response time.
- Tested the system's handling of sensitive legal topics, including racism and terrorism, verifying accurate citations, and its ability to infer correct answers from information only implied in the source text rather than explicitly stated.

EDUCATION

University of Siegen

Siegen, Germany

M.Sc. in Human-Computer Interaction

2026 – 2028

Coursework: HCI Analysis & Evaluation, User Experience Design, Ubiquitous Computing, Usable Security, Psychology

Ghulam Ishaq Khan Institute (GIKI)

Topi, Pakistan

B.Sc. in Artificial Intelligence, CGPA 3.47 / 4.0

2021 – 2025

Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Engineering, Data Science, Algorithms

PUBLICATIONS

VGG Models in Image Captioning: Which Architecture Delivers Better Descriptions?

ICOSST'24, IEEE Xplore · First & Corresponding Author

2024

Encoder-decoder captioning on Flickr8k. BLEU-1 of **0.541 (VGG16)** vs **0.533 (VGG19)**.

Influencing Factors in Facial Recognition and Verification: A Siamese Neural Network Hyperparameter Study

ICECT'24, IEEE Xplore · First Author

2024

Adam with binary cross-entropy reached **99.8% recall** and **99.7% precision** on LFW plus a custom negative set.

The YOLOv8 Edge: Harnessing Custom Datasets for Superior Real-Time Detection

ICET'23, IEEE Xplore · Co-Author

2023

Custom-trained YOLOv8 detector reaching **mAP50 of 0.864** and mAP50-95 of 0.758.

PROJECTS

S.A.R.A, Surgical Assistant Robotic Arm | YOLOv8, Whisper, FLAN-T5, Python, OpenCV

- Bachelor's Final Year Project. Trained a **YOLOv8 tool-detection model** on ~5,000 self-annotated images, using augmentation to counter overfitting.
- Built the voice-command pipeline (**Whisper** for speech-to-text, **FLAN-T5** for intent parsing) that correctly resolved **negated and multi-part commands** rather than relying on keyword matching.

Diabetic Retinopathy Detection via Transfer Learning | TensorFlow, Keras, CNNs

- Benchmarked five CNN architectures (VGG16, ResNet50, InceptionV3, MobileNetV2, EfficientNetB0) for early detection from retinal images across multiple severity levels.

Fashion-MNIST GAN | TensorFlow, Keras, Generative Models

- Implemented a generative adversarial network from scratch, tuning the generator and discriminator balance to stabilise adversarial training and synthesise clothing images.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C/C++, Dart

Generative AI & LLMs: RAG, agentic AI & multi-agent systems, prompt engineering, LLM evaluation & hallucination detection (BLEU/ROUGE), fine-tuning (LoRA, PEFT), embeddings, local & quantized models

Frameworks & Libraries: LangChain, Hugging Face Transformers, Sentence-Transformers, PyTorch, TensorFlow, Keras, Ollama, CTransformers, Whisper, OpenCV, Pandas, Scikit-learn

Tools & Infrastructure: n8n, REST API integration, Docker, Git/GitHub, Linux, PostgreSQL, Jupyter, Claude Code

Vector Databases: ChromaDB, Supabase, Qdrant

Language Skills: English (C1), German (A2)

CERTIFICATIONS

Generative AI: Finetuning Large Language Models (Lamini) · LLMOps (DeepLearning.AI) · Foundational Generative AI (iNeuron.ai) · GenAI in Networking Bootcamp (Aviz Networks)

AWARDS & LEADERSHIP

- **Mitacs Globalink Research Fellowship 2025:** fully funded international research award, University of Regina, Canada.
- **Selected among 32 of 15,000+ applicants** for the SummerSpark Internship, PTCL & Ufone 4G.
- Dean's Honor Roll, 5× during bachelor's studies.
- **Winner, MLSA Hackathon**, GIKI.
- Microsoft Imagine Cup, Regional Finalist.
- **General Secretary, IEEE GIKI Student Branch** (4 years, progressing from volunteer): organised technical events and workshops.
- Huawei Cloud Campus Coordinator for AI, sole AI representative at GIKI.
- Core team member, Google Developer Student Club (GDSC), GIKI.